

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Baseline Model 1 : Sentiment Analysis using TF-IDF and Logistic Regression

1. Model Overview

Baseline Model 1 uses a traditional machine learning approach for sentiment classification. It combines TF-IDF vectorization for feature extraction with a Logistic Regression classifier to categorize text into Positive, Negative, and Neutral sentiments.

2. Feature Representation

The model uses TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual input into numerical vectors. This representation captures the importance of words based on their frequency and uniqueness, enabling the model to identify key terms that influence sentiment.

3. Model Architecture – Logistic Regression

Logistic Regression is a linear classification algorithm that predicts probabilities for each class using a logistic (sigmoid/softmax) function.

✓ Working Principle:

- Takes numerical feature vectors as input
- Computes weighted sum of features
- Applies activation function to produce probabilities
- Final class is selected based on highest probability

4. Training Process

The model is trained using labeled data where:

- Input → TF-IDF vectors
- Output → sentiment class

During training:

- The model learns weights for each feature
- These weights represent how strongly each word contributes to a particular sentiment

5. Model Performance

The Logistic Regression model achieved approximately **75–76% accuracy**.

✓ Observations:

- Performs well for **clear positive and negative sentiments**
- Struggles with **neutral or ambiguous text**
- Performance is influenced by feature quality

6. Advantages of the Model

- Simple and fast to train
- Works well with high-dimensional text data
- Easy to interpret
- Requires less computational power

7. Limitations of the Model

- Cannot capture **context or word order**
- Treats text as independent features (bag-of-words approach)
- Limited understanding of complex language patterns
- Not suitable for capturing deep semantic meaning

Baseline Model 1 demonstrates a fundamental approach to sentiment classification using TF-IDF and Logistic Regression. While effective for basic text analysis, it lacks the ability to understand contextual relationships, which motivates the use of more advanced deep learning models in later stages.

RESULT

Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.76	0.69	0.73	1174
Neutral	0.38	0.08	0.14	476
Positive	0.78	0.94	0.85	2346

Overall Performance

Metric	Value
Accuracy	0.76
Macro Avg F1	0.57
Weighted Avg F1	0.73

Confusion Matrix

